

Software Requirements Specification

MSBON Fraud-Sensitive Transcript Verification System

Version 1.0 | Team Nexus | University of Southern Mississippi | CSC 424 Capstone

Revision History

Table of Contents

1. Introduction

- 1.1 Purpose
- 1.2 Scope
- 1.3 Definitions, Acronyms, and Abbreviations
- 1.4 References
- 1.5 Document Overview

2. Overall Description

- 2.1 Product Perspective
- 2.2 Product Functions
- 2.3 User Classes and Characteristics
- 2.4 Operating Environment
- 2.5 Design and Implementation Constraints
- 2.6 Assumptions and Dependencies

3. External Interface Requirements

- 3.1 User Interfaces
- 3.2 Hardware Interfaces
- 3.3 Software Interfaces
- 3.4 Communications Interfaces

4. Functional Requirements

- 4.1 Authentication and Authorization (F1)
- 4.2 Transcript Upload (F2)
- 4.3 Document Extraction (F3)
- 4.4 Verification (Rule Engine) (F4)
- 4.5 Staff Review and Decision (F5)
- 4.6 Audit Log (F6)
- 4.7 Program Catalog Management (F7)

5. Non-Functional Requirements

- 5.1 Performance (NFR-PERF)
- 5.2 Security (NFR-SEC)
- 5.3 Reliability and Availability (NFR-REL)
- 5.4 Usability (NFR-USE)
- 5.5 Maintainability (NFR-MAINT)
- 5.6 Portability (NFR-PORT)
- 5.7 Privacy (NFR-PRIV)

6. Use Cases

- 6.1 Use Case Diagram
- 6.2 UC-01 — Authenticate
- 6.3 UC-02 — Upload Transcript
- 6.4 UC-03 — Review Flagged Transcript
- 6.5 UC-04 — Browse Transcript List
- 6.6 UC-05 — Export Audit Log
- 6.7 UC-06 — Audit Every State Change (System)
- 6.8 UC-07 — Manage Program Catalog
- 6.9 Sequence Diagram — End-to-End Verification

7. Data Requirements

- 7.1 Logical Data Model (Entity-Relationship)
- 7.2 LLM-Returned JSON Schema (Extracted Data)
- 7.3 Transcript State Machine
- 7.4 Data Retention
- 8. System Constraints and Compliance
- 9. Acceptance Criteria and Traceability Matrix
 - 9.1 Acceptance Criteria (Customer-Facing)
 - 9.2 Functional Requirement → Use Case → Test Traceability
 - 9.3 Demo Script (recorded presentation)
- 10. Appendix
 - 10.1 Mapping to CSC 424 Capstone Deliverables
 - 10.2 Flagging Rules Reference (canonical list — same as SDD §5.4)
 - 10.3 Glossary of Status Codes
 - 10.4 Demo Script Detailed Steps
 - 10.5 Open Issues / Future Work

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Revision History

Version	Date	Author	Description
0.1	2026-05-12	Team Nexus	Initial draft of capstone SRS
1.0	2026-05-12	Team Nexus	Baseline SRS — PoC scope, IEEE 830 structure

Table of Contents

- 1. [Introduction](#)
- 2. [Overall Description](#)
- 3. [External Interface Requirements](#)
- 4. [Functional Requirements](#)
- 5. [Non-Functional Requirements](#)
- 6. [Use Cases](#)
- 7. [Data Requirements](#)
- 8. [System Constraints and Compliance](#)
- 9. [Acceptance Criteria and Traceability Matrix](#)
- 10. [Appendix](#)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional, non-functional, interface, and data requirements for the **MSBON Fraud-Sensitive Transcript Verification System** — a Proof of Concept (PoC) developed by Team Nexus for the Mississippi State Board of Nursing (MSBON) under the CSC 424 Senior Capstone at The University of Southern Mississippi. This document is the binding contract between the development team and the customer (MSBON). It tells the customer exactly **what** the delivered system will do; the companion Software Design Document (SDD) explains **how** the system implements these requirements.

This SRS conforms to IEEE Std 830-1998 in spirit, adapted for an academic capstone deliverable and a single-cycle PoC release. It is the authoritative input to test plans, acceptance review, and the end-of-semester demonstration.

1.2 Scope

1.2.1 Product Identification

The product is named **MSBON Transcript Verification System**, version **1.0.0-poc**. It is delivered as a self-contained web application consisting of a FastAPI HTTP backend (Python 3.11+) and a React 18 single-page frontend (TypeScript + Vite + Tailwind CSS), backed by SQLite for the PoC.

1.2.2 What the Product Does

The system automates the **first-pass review** of nursing school transcripts submitted to MSBON for licensure applications. Specifically it shall:

1. **Ingest** transcript documents (PDF or scanned image) uploaded by authorized MSBON staff.
2. **Extract** structured data from each document using a two-stage pipeline: deterministic OCR/text extraction (pdfplumber + Tesseract) followed by a Large Language Model (Google Gemini gemini-2.0-flash) that produces normalized JSON.
3. **Verify** the extracted data against twelve transparent, rule-based fraud and compliance checks across five categories: Graduation, Accreditation, Course Completion, Fraud Indicators, and Formatting.
4. **Surface** every flag with a human-readable explanation, source excerpt, and severity to the reviewing staff member.
5. **Capture** the staff member's decision — *Confirmed*, *Overridden*, or *Needs More Information* — along with a free-text justification when required.
6. **Persist** every state change in a tamper-evident audit log that is exportable as CSV or JSON for compliance review.

1.2.3 What the Product Does NOT Do (Out of Scope)

#	Out of Scope	Rationale
1	Issue, deny, or revoke a nursing license	All licensure decisions remain with human MSBON staff per board statute.
2	Process foreign (non-U.S.) transcripts	Foreign credential evaluation is delegated to CGFNS in the production process.
3	Communicate with applicants	The system has no applicant-facing surface in this PoC.
4	Long-term storage of original transcript files	Source files are deleted after configurable retention (default 24 h).
5	Integrate with MSBON's production licensing database	PoC is air-gapped from production data.
6	Provide HIPAA/PHI-grade encryption-at-rest	Field-level encryption planned for Phase D infrastructure work.

1.2.4 Benefits and Objectives

Benefit	Measurable Objective
Reduce manual transcript review time	First-pass triage in ≤ 30 s per transcript at 95th percentile
Increase fraud detection sensitivity	≥ 90 % flag accuracy against the synthetic test transcript set
Provide complete audit trail	100 % of state changes produce an audit entry; 0 audit gaps
Preserve human authority	100 % of OVERRIDDEN decisions store a justification
Make decisions explainable	Every flag carries rule ID + source excerpt + plain-language explanation

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
MSBON / MSBN	Mississippi State Board of Nursing — the customer
PoC	Proof of Concept — capstone deliverable, not production
OCR	Optical Character Recognition (Tesseract via pytesseract)
LLM	Large Language Model (Google Gemini gemini-2.0-flash)
SRS / SRD	Software Requirements Specification / Software Requirements Document
SDD	Software Design Document — companion document

Term	Definition
ACEN	Accreditation Commission for Education in Nursing
CCNE	Commission on Collegiate Nursing Education
ADN / BSN / MSN	Associate / Bachelor / Master of Science in Nursing
API	Application Programming Interface
REST	Representational State Transfer
RBAC	Role-Based Access Control
JWT	JSON Web Token (RFC 7519)
ORM	Object-Relational Mapper
SPA	Single Page Application
FERPA	Family Educational Rights and Privacy Act
WCAG	Web Content Accessibility Guidelines
Operation Nightingale	2023 federal investigation into fraudulent nursing diplomas
Flag	A finding produced by a verification rule (rule_id, severity, message, source excerpt)
Rule	A deterministic predicate evaluating extracted transcript data
Reviewer	An MSBON staff member with the reviewer role
Decision	One of CONFIRMED, OVERRIDDEN, or NEEDS_MORE_INFO

1.4 References

#	Reference
R-1	MSBON PoC Scope Document, January 7 2026
R-2	MSBON Project Concept, Team Nexus, February 27 2026
R-3	Software Design Document (SDD), Team Nexus, v1.0 — <code>Software_Design_Document.md</code>
R-4	Phase 1 Audit Report, 2026-03-29 — <code>msbon-app/docs/PHASE1_AUDIT_REPORT.md</code>
R-5	Phase 2 Implementation Blueprint, 2026-03-29 — <code>msbon-app/docs/PHASE2_BLUEPRINT.md</code>
R-6	IEEE Std 830-1998, <i>Recommended Practice for Software Requirements Specifications</i>
R-7	OWASP Top 10 (2021)
R-8	NIST SP 800-63B (digital identity guidelines)
R-9	WCAG 2.1 Level AA
R-10	ACEN and CCNE accreditation standards (publicly published)
R-11	FastAPI documentation, https://fastapi.tiangolo.com/
R-12	Google Gemini API documentation, https://ai.google.dev/
R-13	Operation Nightingale federal indictment, U.S. Attorney for the Southern District of Florida, 2023

1.5 Document Overview

Section 2 paints the system in broad strokes: where it sits in the MSBON workflow, who uses it, and what constraints it operates under. Section 3 enumerates every interface the system presents — to users, to other software, and to the network. Section 4 contains the catalog of functional requirements (REQ-FR-***), each numbered, prioritized, and traceable to a use case. Section 5 covers the non-functional requirements (performance, security, reliability, usability). Section 6 captures the primary use cases as actor flows. Section 7 specifies the data model. Section 8 lists external constraints and compliance obligations. Section 9 defines the acceptance criteria and a traceability matrix linking every requirement to a test or demo step. Section 10 holds the appendices.

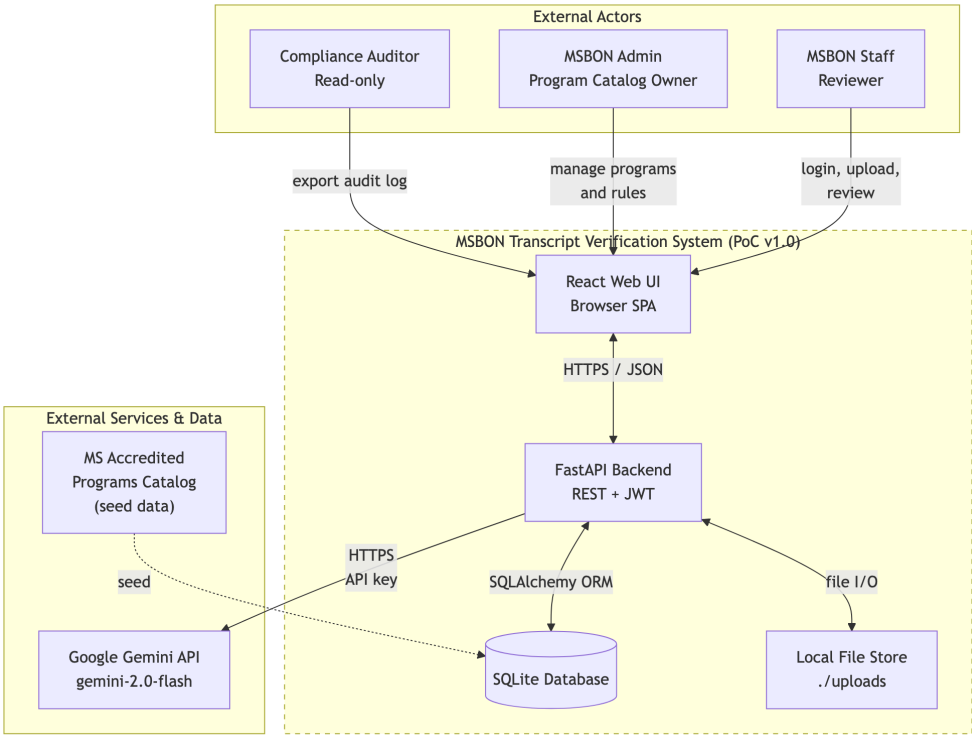
2. Overall Description

2.1 Product Perspective

The MSBON Transcript Verification System is a **new, self-contained product** with no dependency on existing MSBON information systems. It is intended to be operated on a workstation or private server inside the MSBON network. The system *consumes* one external service (the Google Gemini API over HTTPS) and *produces* output that is exclusively read by human MSBON staff. It is not a component of a larger system in this PoC; integration with the production licensing database is explicitly deferred to a follow-on engagement.

2.1.1 System Context Diagram

The diagram below shows the system boundary, the external actors, and the external services. Anything inside the dashed box is delivered by Team Nexus; everything outside is supplied or operated by another party.



Srd Context Diagram

2.2 Product Functions

The system implements seven top-level capabilities. Each is broken down into specific functional requirements in Section 4.

#	Capability	Primary Actor	FR Range
F1	Authenticate and authorize staff members (RBAC)	Reviewer / Admin	FR-001 – FR-010
F2	Upload nursing transcripts (PDF or image)	Reviewer / Admin	FR-020 – FR-029
F3	Extract structured data via OCR + LLM	System (background task)	FR-030 – FR-039
F4	Apply 12 transparent verification rules	System	FR-040 – FR-052
F5	Review, annotate, and decide on flagged transcripts	Reviewer	FR-060 – FR-074
F6	Maintain tamper-evident audit log	System (write-only)	FR-080 – FR-089
F7	Manage accredited program catalog	Admin	FR-090 – FR-099

2.3 User Classes and Characteristics

User Class	Count	Frequency	Technical Skill	Privileges (RBAC)
Reviewer (MSBON licensure staff)	~5–15	Daily	Comfortable with web apps; not technical	Upload, list, review, annotate; cannot manage rules or programs
Admin (MSBON IT / program catalog owner)	1–2	Weekly	Technically literate	All Reviewer rights + manage programs, rules, view system status
Viewer (Compliance auditor)	1–3	Quarterly	Audit-focused	Read-only access to transcripts and audit log; no upload, no review
Developer (Team Nexus, capstone team)	5	During development only	Full-stack	Direct DB / filesystem access in dev environment only

The **Reviewer** is the primary user — interface design (font size, contrast, button placement, plain-language flag explanations) is optimized for this class first.

2.4 Operating Environment

Element	Specification
Server OS	macOS 13+, Ubuntu 22.04 LTS, or Windows 11 (PoC supports any platform with Python 3.11)
Server Runtime	Python 3.11+, Node.js 18+ (build only)
Server Hardware	4 GB RAM minimum, 10 GB free disk for uploads + DB; no GPU required
Database	SQLite 3.39+ (PoC); upgrade path to PostgreSQL 15+ documented in SDD §11.2
Client	Any evergreen browser supporting ES2022 (Chrome 110+, Firefox 110+, Safari 16+, Edge 110+)
Client Hardware	Standard office workstation; 1366×768 minimum display
Network	Local or private network; outbound HTTPS to <code>generativelanguage.googleapis.com</code> for LLM calls
External Service	Google Gemini API (active API key required)

2.5 Design and Implementation Constraints

#	Constraint	Origin
C-1	Backend language must be Python (FastAPI)	Course tooling alignment + team skill
C-2	Frontend must be React + TypeScript	Course tooling alignment
C-3	LLM provider must be Google Gemini for the PoC	Customer-supplied API access
C-4	All AI outputs are advisory; no automated licensure decisions	MSBON board policy
C-5	Only synthetic or de-identified transcript data may be used during development	FERPA / privacy
C-6	All source code, scripts, and documents under git version control	CSC 424 capstone tooling requirement
C-7	Mermaid is the canonical diagramming notation for both SRS and SDD	Team standard; CI-renderable
C-8	The product must be demonstrable end-to-end in a recorded video at semester end	CSC 424 presentation deliverable
C-9	No long-term retention of raw transcript files	Privacy + storage cost

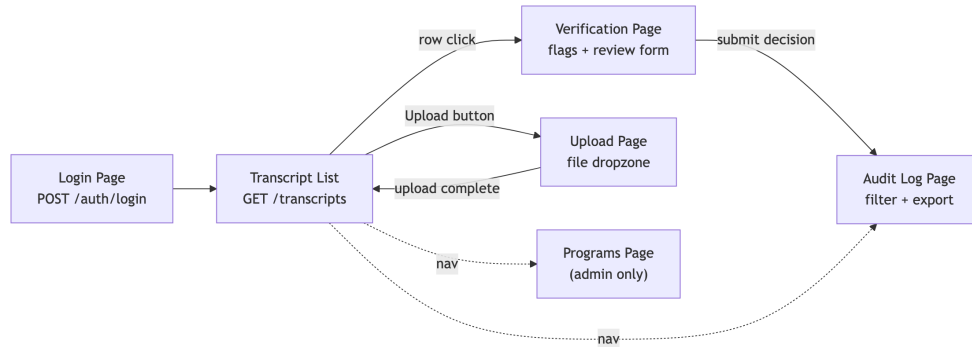
2.6 Assumptions and Dependencies

#	Assumption / Dependency	If Violated
A-1	A valid Google Gemini API key is provisioned in GEMINI_API_KEY	Extraction fails with LLM_UNAVAILABLE; uploads queue but never process
A-2	Tesseract OCR is installed on the host (brew install tesseract / apt install tesseract-ocr)	Image-based transcripts cannot be extracted
A-3	libmagic is installed for MIME validation (brew install libmagic)	File-type detection falls back to extension only — security regression
A-4	The MSBON-approved accredited program list is seeded into the database	ACCR-001 / ACCR-002 rules over-flag every transcript
A-5	Reviewers are trained on what each flag category means	Decision quality degrades; OVERRIDDEN rate spikes
A-6	Network access to generativelanguage.googleapis.com is permitted	Same as A-1

3. External Interface Requirements

3.1 User Interfaces

The system provides a single web-based UI served at `http://<host>:5173` in development (and a static-asset bundle for deployment). The information architecture is summarized below.



Srd Information Architecture

UI requirements:

ID	Requirement
UI-001	The UI shall present a top-level navigation with links: Transcripts, Upload, Programs (admin-only), Audit Log, and a session indicator (logged-in user, role, log-out).
UI-002	The UI shall be operable on viewports 1366×768 and larger; mobile responsiveness is <i>not</i> required for the PoC.
UI-003	The UI shall conform to WCAG 2.1 Level AA contrast ratios for all text on background.
UI-004	All actionable controls shall be reachable by keyboard navigation (Tab / Shift-Tab / Enter / Space).
UI-005	Status indicators (transcript status, flag severity) shall convey meaning by <i>both</i> color and text/icon — never color alone.
UI-006	All asynchronous operations (upload, processing, submit) shall display either a deterministic progress indicator or a textual “in progress” state within 200 ms of trigger.
UI-007	Server-side errors shall be surfaced to the user with the human-readable <code>error.message</code> field; raw stack traces shall never be displayed.
UI-008	The UI shall obey the design tokens defined in <code>frontend/docs/DESIGN_TOKENS.md</code> (no hardcoded colors).

3.2 Hardware Interfaces

The system imposes no special hardware requirements beyond the operating environment in §2.4. There is no integration with scanners, ID readers, biometric devices, or printers in the PoC scope.

3.3 Software Interfaces

Interface	Direction	Protocol	Auth	Note
Browser → Backend REST API	Bidirectional	HTTPS, JSON	JWT bearer (header Authorization: Bearer <token>)	All endpoints under /a
Backend → Google Gemini	Outbound	HTTPS	API key (env var GEMINI_API_KEY)	Endpoint generativelanguage gemini-2.0-flash m
Backend → SQLite	Local	SQLAlchemy ORM	Filesystem permissions	DSN sqlite:///./da Alembic-managed sche
Backend → Local file store	Local	POSIX file I/O	Filesystem permissions	Files written as {uuid} ./uploads, MIME-val libmagic
Backend → Tesseract	Subprocess	pytesseract shell-out	none	Used by DocumentExt transcripts

3.4 Communications Interfaces

ID	Requirement
COM-001	All HTTP traffic in production deployment shall use TLS 1.2 or higher. (PoC dev mode permits plain HTTP on 127.0.0.1.)
COM-002	The backend shall accept CORS pre-flight requests only from origins listed in CORS_ORIGINS (default http://localhost:5173).
COM-003	The backend shall expose its OpenAPI 3.1 schema at /openapi.json and an interactive Swagger UI at /docs.
COM-004	All error responses shall conform to a uniform envelope {"error": {"code", "message", "detail", "timestamp"}}.
COM-005	The backend shall propagate or generate an X-Correlation-ID header on every request and include it in every log line for that request.

4. Functional Requirements

Each requirement is identified REQ-FR-NNN, has a **Priority** (Must / Should / May per MoSCoW), and traces to one or more use cases in §6.

Must = required for capstone demo and acceptance. *Should* = scheduled for the PoC but acceptable to defer with justification. *May* = stretch goal; included only if time permits.

4.1 Authentication and Authorization (F1)

ID	Requirement	Priority	Trace
FR-001	The system shall authenticate users via a POST /api/v1/auth/login endpoint accepting staff_id and password, returning a signed JWT with payload {sub, role, exp}.	Must	UC-01
FR-002	The system shall reject login attempts with invalid credentials with HTTP 401 and the error envelope from COM-004.	Must	UC-01
FR-003	Issued JWTs shall expire after a configurable interval (default 480 minutes = 8-hour shift).	Must	UC-01
FR-004	Every protected endpoint shall require a valid JWT in the Authorization: Bearer header.	Must	All UCs
FR-005	The system shall implement role-based access control with three roles: admin, reviewer, viewer.	Must	UC-01–UC-07
FR-006	The system shall enforce permissions per role as defined in app/auth/permissions.py (Admin: all; Reviewer: upload + review; Viewer: read-only).	Must	UC-01–UC-07
FR-007	The system shall reject requests carrying expired or tampered tokens with HTTP 401.	Must	All UCs
FR-008	Client-supplied identity headers (X-Staff-ID, X-Staff-Role) shall <i>never</i> be used for authorization decisions.	Must	UC-01
FR-009	The system shall log every successful login and every authorization failure to the audit log (AUTH_LOGIN, AUTH_DENIED).	Must	UC-06
FR-010	The system <i>should</i> redirect a user to the login page upon	Should	UC-01

ID	Requirement	Priority	Trace
	receiving an HTTP 401 from any API call.		

4.2 Transcript Upload (F2)

ID	Requirement	Priority	Trace
FR-020	The system shall accept transcript uploads via POST /api/v1/transcripts as multipart/form-data with a single file part.	Must	UC-02
FR-021	Accepted file types shall be: application/pdf, image/png, image/jpeg, image/tiff.	Must	UC-02
FR-022	File type shall be validated by inspecting magic bytes (via python-magic), <i>not</i> by trusting the filename extension.	Must	UC-02
FR-023	Maximum accepted file size shall be configurable (MAX_FILE_SIZE_MB, default 10 MB); larger files shall be rejected with HTTP 413.	Must	UC-02
FR-024	Uploaded files shall be stored under UPLOAD_DIR with a randomized name {uuid}.{ext}; the original filename shall be preserved as metadata only.	Must	UC-02
FR-025	A successful upload shall return HTTP 202 with a JSON body containing {transcript_id, status: "QUEUED"} and trigger an asynchronous extraction task.	Must	UC-02
FR-026	The system shall create an audit entry (TRANSCRIPT_UPLOADED) for every successful upload, recording transcript_id, staff_id, filename, file_size, and timestamp.	Must	UC-06
FR-027	If validation fails, no file shall be persisted to disk and no DB row shall be created.	Must	UC-02
FR-028	The system <i>should</i> delete uploaded files after FILE_RETENTION_HOURS	Should	UC-02

ID	Requirement	Priority	Trace
	(default 24 h) once processing is complete.		
FR-029	The UI <i>should</i> display a deterministic upload progress bar driven by the XHR onprogress event.	Should	UC-02

4.3 Document Extraction (F3)

ID	Requirement	Priority	Trace
FR-030	The system shall extract raw text from PDF transcripts using pdfplumber.	Must	UC-02
FR-031	The system shall extract raw text from image transcripts using Tesseract via pytesseract.	Must	UC-02
FR-032	Image inputs shall be preprocessed (grayscale conversion, contrast enhancement) before OCR.	Should	UC-02
FR-033	The extracted raw text shall be passed to the Google Gemini LLM with a deterministic prompt template that returns JSON conforming to the schema in §7.2.	Must	UC-02
FR-034	The LLM adapter shall validate the response against the JSON schema; non-conforming responses shall be retried (≤ 3 attempts) with exponential back-off (1 s, 2 s, 4 s).	Must	UC-02
FR-035	If all retries fail, the transcript status shall be set to FAILED and an audit entry EXTRACTION_FAILED shall be created with the underlying error code.	Must	UC-02, UC-06
FR-036	The system shall persist extracted data as a ExtractedData row linked one-to-one with Transcript.	Must	UC-02
FR-037	Upon successful extraction, the transcript status shall transition QUEUED → PROCESSING → EXTRACTED.	Must	UC-02

ID	Requirement	Priority	Trace
FR-038	The Gemini API key shall be read from environment variables only; it shall never appear in logs, error messages, or HTTP responses.	Must	NFR-SEC
FR-039	The system <i>may</i> support pluggable LLM providers via an LLMAPI interface for future replacement.	May	—

4.4 Verification (Rule Engine) (F4)

ID	Requirement	Priority	Trace
FR-040	After successful extraction, the system shall automatically invoke the Rule Engine over the extracted JSON.	Must	UC-02
FR-041	The Rule Engine shall load the active rule set from the <code>flagging_rules</code> table (only rows with <code>is_active=true</code>).	Must	UC-02
FR-042	The system shall implement and ship the following twelve rules — see §10.2 for the canonical reference: GRAD-001, GRAD-002, ACCR-001, ACCR-002, COUR-001, COUR-002, FRAU-001, FRAU-002, FRAU-003, FORM-001, FORM-002, FORM-003.	Must	UC-02, UC-03
FR-043	Every emitted flag shall include <code>rule_id</code> , <code>severity</code> (HIGH / MEDIUM / LOW), <code>message</code> , <code>source_excerpt</code> (verbatim text from the transcript), and <code>explanation</code> (≥ 1 plain-language sentence).	Must	UC-03
FR-044	Verification results shall be persisted as <code>VerificationFlag</code> rows linked to the parent transcript.	Must	UC-02
FR-045	Upon completion, the transcript status shall transition EXTRACTED → AWAITING_REVIEW if any flags exist, or EXTRACTED → CLEARED if zero flags.	Must	UC-02
FR-046	An audit entry <code>VERIFICATION_COMPLETED</code> shall be written with the flag count and rule IDs that fired.	Must	UC-06

ID	Requirement	Priority	Trace
FR-047	The system shall allow the active rule set to be modified by an admin user via PATCH <code>/api/v1/rules/{rule_id}</code> (toggle <code>is_active</code>).	Should	UC-07
FR-048	Rule evaluation shall be deterministic: identical extracted JSON + identical active rule set shall always yield identical flags.	Must	NFR-REL
FR-049	A failure inside any single rule shall not prevent other rules from running; the failed rule shall produce a flag of severity HIGH with <code>rule_id="RULE_INTERNAL_ERROR"</code> .	Should	NFR-REL
FR-050	The system <i>should</i> allow admins to add custom rules via configuration without code changes (post-PoC).	May	—
FR-051	The Rule Engine shall complete evaluation of a single transcript within 2 s on PoC reference hardware (M1 MacBook).	Must	NFR-PERF
FR-052	Accreditation rules shall consult the <code>accredited_programs</code> catalog (institution name, body, <code>is_active</code>); rules shall <i>not</i> hardcode program names.	Must	UC-07

4.5 Staff Review and Decision (F5)

ID	Requirement	Priority	Trace
FR-060	The system shall provide GET <code>/api/v1/transcripts</code> returning a paginated list filterable by status.	Must	UC-04
FR-061	The system shall provide GET <code>/api/v1/transcripts/{id}</code> returning the transcript, its extracted data, and all flags.	Must	UC-03
FR-062	The Verification page shall display flags grouped by severity (HIGH first, then MEDIUM, then LOW).	Must	UC-03
FR-063	Each flag shall be expandable to show the <code>source_excerpt</code> and <code>explanation</code> inline.	Must	UC-03
FR-064	The Review form shall provide three exclusive decision controls — <code>CONFIRMED</code> , <code>OVERRIDDEN</code> , <code>NEEDS_MORE_INFO</code> — selectable by mouse or keyboard.	Must	UC-03

ID	Requirement	Priority	Trace
FR-065	If OVERRIDDEN is selected, the user shall be required to enter an <code>override_reason</code> of at least 10 characters before submission.	Must	UC-03
FR-066	Submission shall be a single POST <code>/api/v1/transcripts/{id}/reviews</code> carrying <code>{decision, override_reason?, annotation?}</code> .	Must	UC-03
FR-067	Upon successful submission, the transcript status shall transition to REVIEWED and an audit entry REVIEW_SUBMITTED shall be written.	Must	UC-03, UC-06
FR-068	A transcript may be reviewed multiple times; each review shall be persisted as a separate <code>StaffReview</code> row. The most recent decision is the operative one.	Should	UC-03
FR-069	The system shall reject review submissions on transcripts whose status is not in <code>{AWAITING_REVIEW, REVIEWED}</code> with HTTP 409.	Must	UC-03
FR-070	A reviewer with the <code>viewer</code> role shall be able to read flags but shall be denied submission with HTTP 403.	Must	UC-03
FR-071	The system shall record <code>reviewer_id</code> from the JWT sub claim — never from a request body field.	Must	NFR-SEC
FR-072	The UI shall provide a confirmation prompt (modal or inline) before a OVERRIDDEN submission is sent.	Should	UC-03
FR-073	The system shall allow free-text reviewer annotations of up to 2,000 characters per review.	Should	UC-03
FR-074	The system <i>may</i> allow attaching supporting documents to a review (post-PoC).	May	—

4.6 Audit Log (F6)

ID	Requirement	Priority	Trace
FR-080	The system shall create an <code>AuditLog</code> entry for every state-changing operation (upload, extraction success/failure, verification, review, login, role change).	Must	UC-06
FR-081	Each <code>AuditLog</code> entry shall capture <code>id</code> , <code>timestamp</code> (UTC),	Must	UC-06

ID	Requirement	Priority	Trace
	actor_id, actor_role, action_type, transcript_id (nullable), action_detail (JSON), ip_address, correlation_id.		
FR-082	The AuditRepository shall be write-only : update() and delete() shall raise OperationNotPermittedError.	Must	NFR-SEC
FR-083	The system shall provide GET /api/v1/audit returning paginated audit entries filterable by transcript_id, actor_id, action_type, and date range.	Must	UC-06
FR-084	The system shall provide GET /api/v1/audit/export?format={csv json} returning the full filtered audit set as a downloadable file.	Must	UC-06
FR-085	Audit entries shall <i>never</i> be edited or deleted by any code path; corrections must be issued as new compensating audit entries.	Must	NFR-SEC
FR-086	The action_detail field shall not contain raw transcript text or PII; it shall reference the transcript_id for joinable lookups.	Must	NFR-PRIV
FR-087	Audit entries shall be ordered by timestamp DESC by default in list responses.	Must	UC-06
FR-088	The audit list shall be paginated with default page size 50 and maximum 500.	Must	UC-06
FR-089	The CSV export shall include a header row and shall be quoted per RFC 4180.	Must	UC-06

4.7 Program Catalog Management (F7)

ID	Requirement	Priority	Trace
FR-090	The system shall provide GET /api/v1/programs returning all accredited nursing programs.	Must	UC-07
FR-091	The system shall provide POST /api/v1/programs (admin	Must	UC-07

ID	Requirement	Priority	Trace
	only) creating a new accredited program.		
FR-092	The system shall provide PATCH /api/v1/programs/{id} (admin only) updating fields including is_active.	Must	UC-07
FR-093	Each program record shall capture institution_name, program_name, accreditation_body (ACEN CCNE), accreditation_type (ADN Baccalaureate Master Doctoral), state (2-letter), is_active, accreditation_expires (nullable date).	Must	UC-07
FR-094	The system shall seed the database on first start with the canonical Mississippi accredited program list (see SDD §11.5).	Must	UC-07
FR-095	Program changes shall be captured in the audit log (PROGRAM_CREATED, PROGRAM_UPDATED, PROGRAM_DEACTIVATED).	Must	UC-06, UC-07
FR-096	The system shall validate that accreditation_body ∈ {ACEN, CCNE} and reject other values with HTTP 422.	Must	UC-07
FR-097	The Programs page shall present an inline editor with role-based visibility (Reviewers see read-only; Admins see editable).	Must	UC-07
FR-098	The system <i>should</i> provide a bulk-import endpoint for CSV upload of programs (admin only).	May	—
FR-099	The system <i>should</i> warn at upload time if accreditation_expires is within 90 days.	Should	UC-07

5. Non-Functional Requirements

5.1 Performance (NFR-PERF)

ID	Requirement	Target	Measurement
NFR-PERF-01	A single transcript upload (≤ 10 MB) shall return HTTP 202 within	≤ 1 s @ p95	Manual stopwatch + browser DevTools
NFR-PERF-02	End-to-end processing (upload $\rightarrow$ AWAITING_REVIEW) shall complete within	≤ 30 s @ p95	Background task instrumentation
NFR-PERF-03	The Verification page shall load and render flags within	≤ 1.5 s @ p95 on broadband	Lighthouse / browser perf
NFR-PERF-04	The system shall sustain	≥ 5 concurrent reviewers without degradation	Manual concurrent test
NFR-PERF-05	Audit log queries with date range ≤ 30 days shall return within	≤ 800 ms	Server timing header

5.2 Security (NFR-SEC)

ID	Requirement
NFR-SEC-01	All authentication shall use stateless JWT (HMAC-SHA256) signed with a server-side secret of ≥ 32 bytes.
NFR-SEC-02	The JWT secret (JWT_SECRET) shall be supplied via environment variable; the system shall refuse to start if it is missing or shorter than 32 bytes.
NFR-SEC-03	All external API keys (GEMINI_API_KEY) shall be supplied via environment variable; they shall not be persisted, logged, or returned in responses.
NFR-SEC-04	Passwords (when implemented in Phase B) shall be hashed with bcrypt cost ≥ 12 .
NFR-SEC-05	The system shall mitigate the OWASP Top 10 (2021): Broken Access Control (RBAC), Cryptographic Failures (TLS + secret hygiene), Injection (parameterized SQL via ORM), Insecure Design (threat model in SDD §7.1), Security Misconfiguration (env validation), Vulnerable Components (pip-audit in CI), ID&Auth Failures (JWT expiry), SSRF (no user-supplied URLs reach the LLM call).
NFR-SEC-06	File uploads shall be rejected if the detected MIME type does not match the claimed extension.
NFR-SEC-07	The system shall set X-Content-Type-Options: nosniff, X-Frame-Options: DENY, and Strict-Transport-Security

ID	Requirement
	headers on every response in production.
NFR-SEC-08	All authorization decisions shall come from the JWT payload — never from request bodies, query parameters, or non-Authorization headers.
NFR-SEC-09	The audit log shall be append-only at the application layer (<code>AuditRepository.update/delete</code> raise <code>OperationNotPermittedError</code>).
NFR-SEC-10	Error responses shall not include stack traces, file paths, or framework internals in production.

5.3 Reliability and Availability (NFR-REL)

ID	Requirement
NFR-REL-01	The Gemini LLM adapter shall retry on transient errors (5xx, 429, network timeouts) up to 3 times with exponential back-off; non-transient errors (401, 400) shall fail fast.
NFR-REL-02	A failure in the extraction pipeline shall leave the database in a consistent state — either the full (<code>Transcript</code> , <code>ExtractedData</code> , <code>AuditLog</code>) triple is committed or none of it is.
NFR-REL-03	The system shall recover automatically from a process restart: queued / processing transcripts shall be retried by the next start.
NFR-REL-04	The system shall expose <code>/api/v1/health</code> returning <code>{status, database, version, timestamp}</code> for readiness probes.
NFR-REL-05	Target availability for a hosted deployment: $\geq 99\%$ over a calendar month.

5.4 Usability (NFR-USE)

ID	Requirement
NFR-USE-01	A new reviewer shall be able to complete their first transcript review within 10 minutes given only the in-app text and tooltips.
NFR-USE-02	Every flag explanation shall be readable at a Flesch-Kincaid grade level ≤ 10 .
NFR-USE-03	The system shall comply with WCAG 2.1 Level AA.
NFR-USE-04	All form fields shall have visible labels (no placeholder-only inputs).
NFR-USE-05	All destructive or irreversible actions (e.g., <code>OVERRIDDEN</code> decision submission) shall require explicit confirmation.

5.5 Maintainability (NFR-MAINT)

ID	Requirement
NFR-MAINT-01	All code shall be formatted with <code>black</code> (Python) and <code>prettier</code> (TS/React).

ID	Requirement
NFR-MAINT-02	All Python code shall pass <code>ruff</code> check with the project ruleset.
NFR-MAINT-03	Backend test coverage shall reach $\geq 60\%$ line coverage by end of PoC (Phase C target: 80 %).
NFR-MAINT-04	All schema changes shall be expressed as Alembic migrations under <code>backend/migrations/versions/</code> .
NFR-MAINT-05	Service classes shall depend on repository / adapter interfaces, not concrete implementations, to enable mocking.
NFR-MAINT-06	The OpenAPI schema at <code>/openapi.json</code> shall be the single source of truth for the HTTP contract.

5.6 Portability (NFR-PORT)

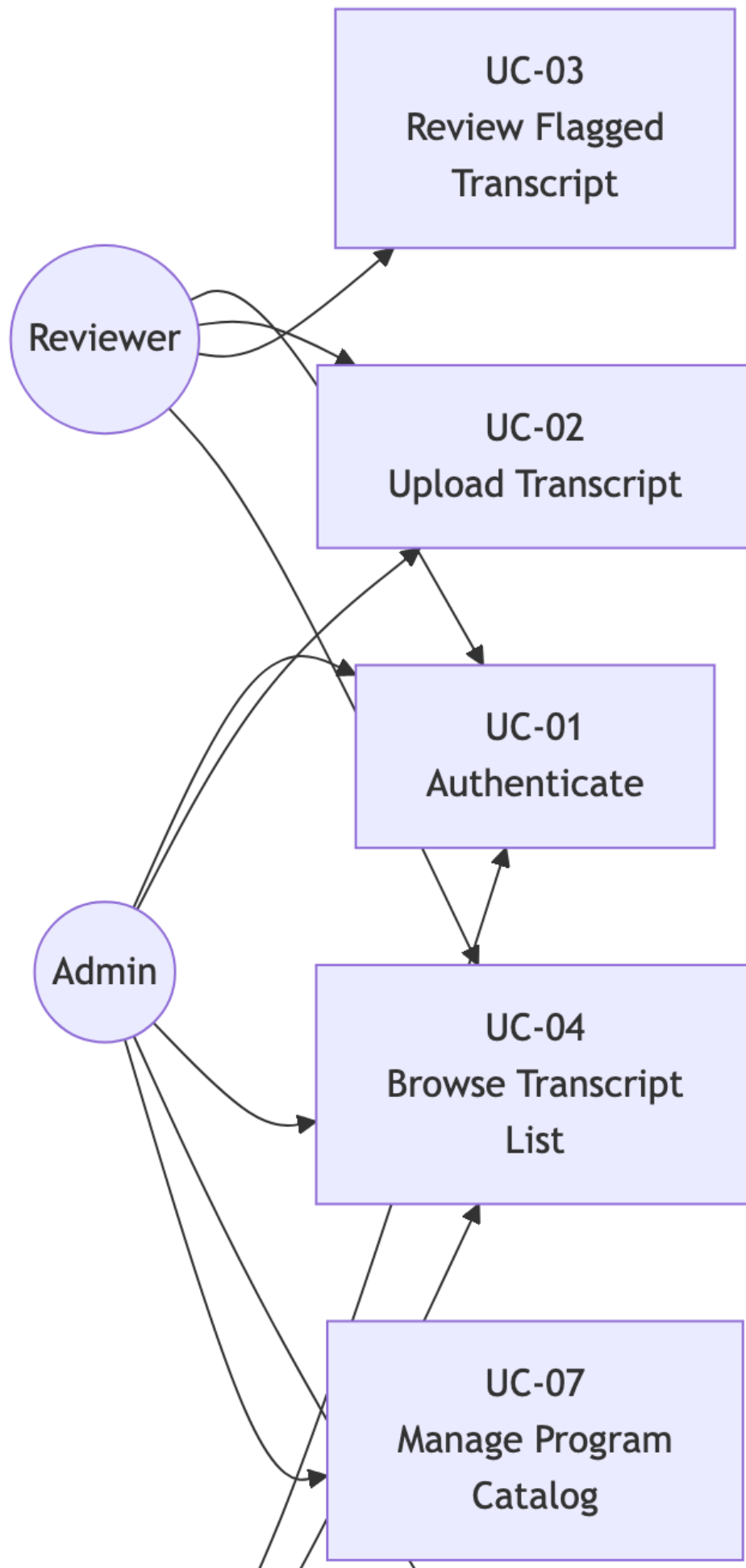
ID	Requirement
NFR-PORT-01	The backend shall run on macOS 13+, Ubuntu 22.04 LTS, and Windows 11 with no source changes.
NFR-PORT-02	The application shall be deployable as a Docker container (<code>docker-compose up</code>).
NFR-PORT-03	The database adapter shall be SQLAlchemy-backed so the SQLite $\rightarrow$ PostgreSQL upgrade requires only a connection-string change.

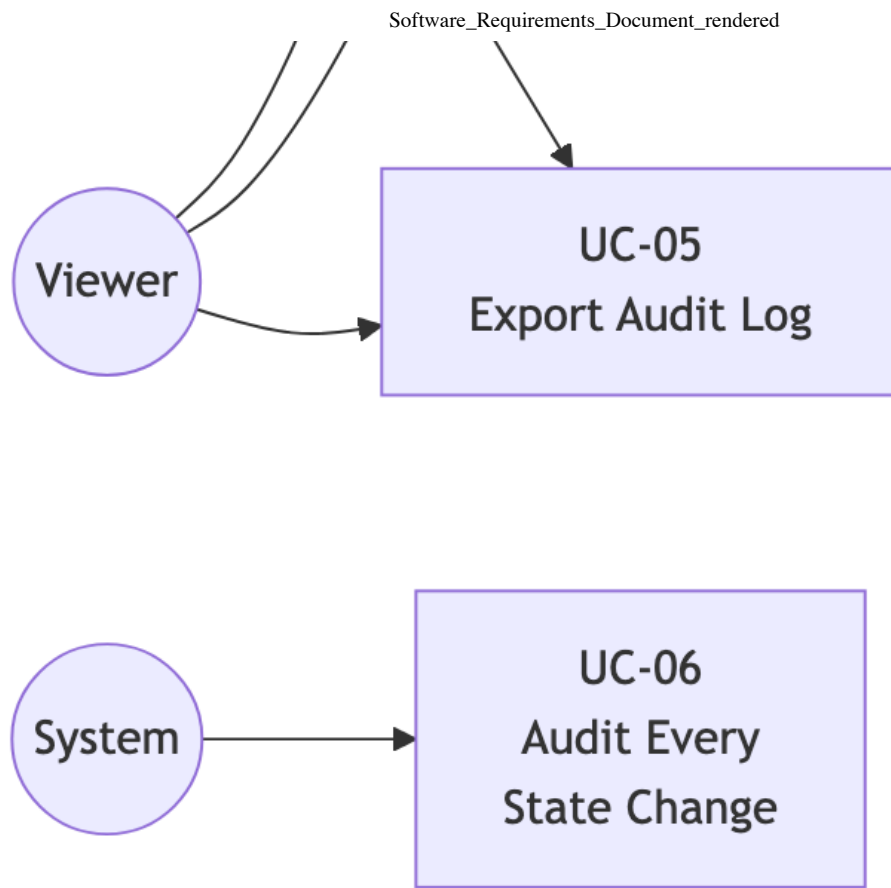
5.7 Privacy (NFR-PRIV)

ID	Requirement
NFR-PRIV-01	Only synthetic / de-identified transcripts shall be used during development and demonstration.
NFR-PRIV-02	Original uploaded files shall be deleted after the configurable retention window (<code>FILE_RETENTION_HOURS</code> , default 24 h).
NFR-PRIV-03	Audit log entries shall not contain raw transcript text — only <code>transcript_id</code> references.
NFR-PRIV-04	The system shall not transmit transcript content to any third party other than the configured LLM provider.
NFR-PRIV-05	The system shall provide an admin operation to permanently purge a transcript and all dependent records (cascades) in support of right-to-erasure requests. (<i>Phase B</i>)

6. Use Cases

6.1 Use Case Diagram





Srd Use Case Diagram

6.2 UC-01 — Authenticate

Primary Actor: Reviewer / Admin / Viewer. **Preconditions:** User has valid credentials provisioned by an Admin. **Trigger:** User opens the application URL.

Main Success Scenario: 1. The system displays the Login page. 2. The user enters `staff_id` and password and clicks *Sign In*. 3. The system validates credentials against the user store. 4. The system issues a JWT with payload `{sub, role, exp}` and returns it to the client. 5. The client stores the token in memory and redirects to the Transcript List. 6. The system writes an `AUTH_LOGIN` audit entry.

Extensions: - 3a. Credentials invalid → return HTTP 401, display “Invalid credentials,” write `AUTH_DENIED` audit entry. - 4a. Token signing fails → return HTTP 500, display generic error.

6.3 UC-02 — Upload Transcript

Primary Actor: Reviewer / Admin. **Preconditions:** User authenticated; Gemini API key configured; Tesseract installed. **Trigger:** User clicks *Upload* on the Transcript List page.

Main Success Scenario: 1. The system displays the Upload page with a drag-and-drop area. 2. The user drops or selects a `.pdf / .png / .jpg / .tiff` file ≤ 10 MB. 3. The client posts the file to `POST /api/v1/transcripts`. 4. The system validates type (magic bytes) and size; persists the file as `{uuid}.{ext}`; creates a Transcript row with status `QUEUED`; returns HTTP 202 with `transcript_id`. 5. A background task picks up the row,

calls DocumentExtractor then LLMAdapter; persists ExtractedData; transitions status QUEUED → PROCESSING → EXTRACTED. 6. The Rule Engine runs over the extracted JSON and persists VerificationFlag rows; status transitions EXTRACTED → AWAITING_REVIEW (or → CLEARED if no flags). 7. The UI polls or websockets-into status updates and surfaces the row in *Awaiting Review*. 8. Each step writes an audit entry per FR-026 / FR-035 / FR-046.

Extensions: - 4a. Invalid file → HTTP 422, no DB row, no file persisted. - 4b. File too large → HTTP 413. - 5a. OCR/extraction fails → status FAILED, EXTRACTION_FAILED audit entry, surfaced as red row in UI. - 5b. LLM unavailable → 3 retries with back-off; if all fail, same as 5a but error code LLM_UNAVAILABLE.

6.4 UC-03 — Review Flagged Transcript

Primary Actor: Reviewer. **Preconditions:** Transcript is in status AWAITING_REVIEW or REVIEWED. **Trigger:** Reviewer clicks the transcript row.

Main Success Scenario: 1. The system loads GET /api/v1/transcripts/{id} returning extracted data + flags. 2. The Verification page renders the transcript summary, then the flag list grouped by severity. 3. The reviewer expands each flag to read source_excerpt + explanation. 4. The reviewer chooses a decision (CONFIRMED / OVERRIDDEN / NEEDS_MORE_INFO). 5. If OVERRIDDEN, the reviewer enters a justification ≥ 10 chars. 6. Optional annotation up to 2 000 chars. 7. The reviewer submits → POST /api/v1/transcripts/{id}/reviews. 8. The system persists the StaffReview, transitions status to REVIEWED, writes REVIEW_SUBMITTED audit entry. 9. UI shows confirmation toast and returns to the list.

Extensions: - 7a. Submission rejected because status was already CLEARED → HTTP 409, no state change. - 7b. Viewer role attempts submission → HTTP 403, no state change.

6.5 UC-04 — Browse Transcript List

Primary Actor: Reviewer / Admin / Viewer. **Trigger:** User opens the Transcript List page.

Main Success Scenario: 1. System loads GET /api/v1/transcripts?status=... &page=.... 2. The page renders a paginated table with columns: filename, uploaded_at, status, flag count, last action. 3. Filter chips allow narrowing by status; sort by uploaded_at. 4. Clicking a row navigates to the Verification page (UC-03).

6.6 UC-05 — Export Audit Log

Primary Actor: Admin / Viewer (compliance auditor). **Trigger:** User clicks *Export CSV* on the Audit Log page.

Main Success Scenario: 1. User opens Audit Log; sets filters (date range, action type, transcript ID, actor). 2. User clicks *Export CSV* (or JSON). 3. System queries the matching audit rows and streams a text/csv (or application/json) response with Content-Disposition: attachment. 4. Browser saves the file audit_export_{timestamp}.csv.

6.7 UC-06 — Audit Every State Change (System)

Primary Actor: System (no human trigger). **Trigger:** Any state-changing handler in the API or services layer.

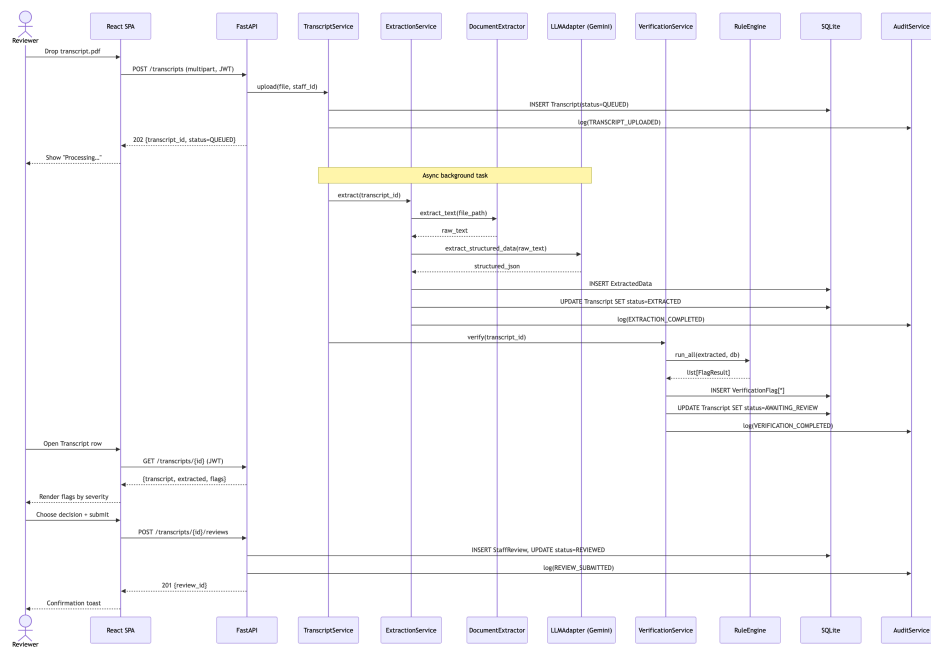
Main Success Scenario: Each handler calls `AuditService.log(action_type, transcript_id?, actor, detail)` inside the same database transaction as the state change, ensuring atomicity. The audit row is uneditable thereafter (NFR-SEC-09).

6.8 UC-07 — Manage Program Catalog

Primary Actor: Admin. **Trigger:** Admin opens the Programs page.

Main Success Scenario: 1. System loads `GET /api/v1/programs`. 2. Admin clicks *Add Program*; fills the form (institution, program, body, type, state, expires); submits. 3. System validates `body ∈ {ACEN, CCNE}` and persists. 4. System writes `PROGRAM_CREATED` audit entry. 5. New program is immediately consulted by ACCR-001 / ACCR-002 on subsequent verifications.

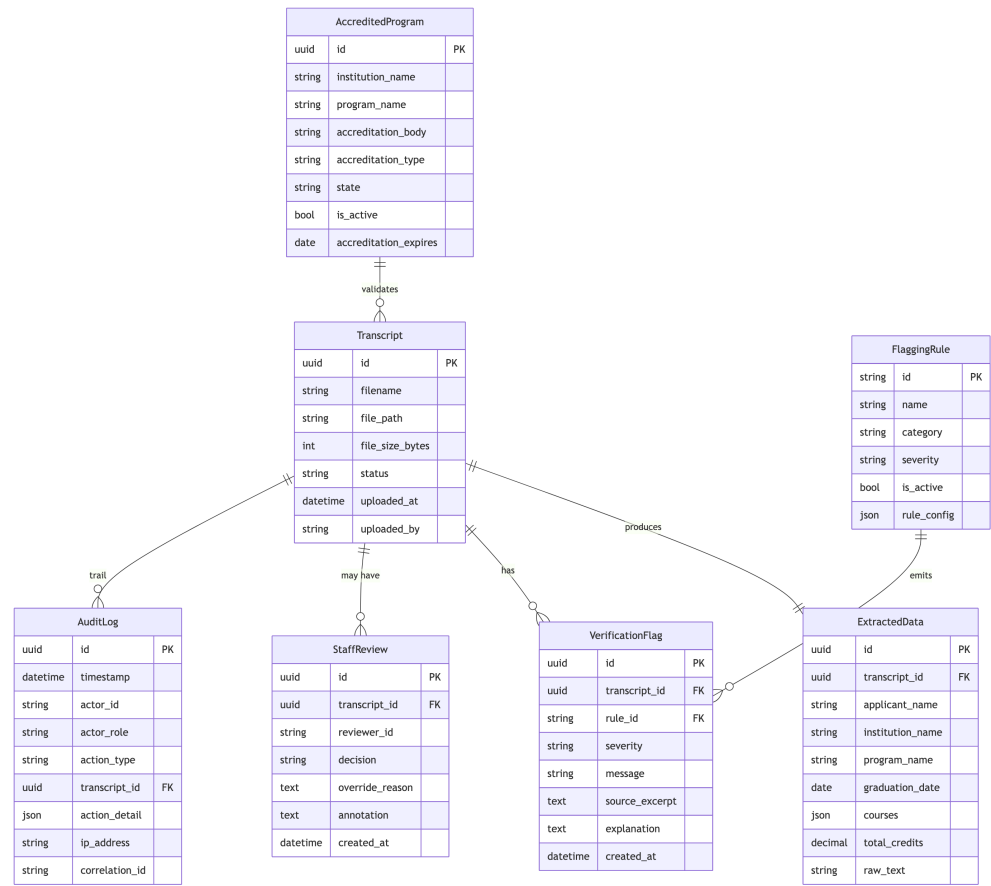
6.9 Sequence Diagram — End-to-End Verification



Srd Sequence E2E

7. Data Requirements

7.1 Logical Data Model (Entity-Relationship)

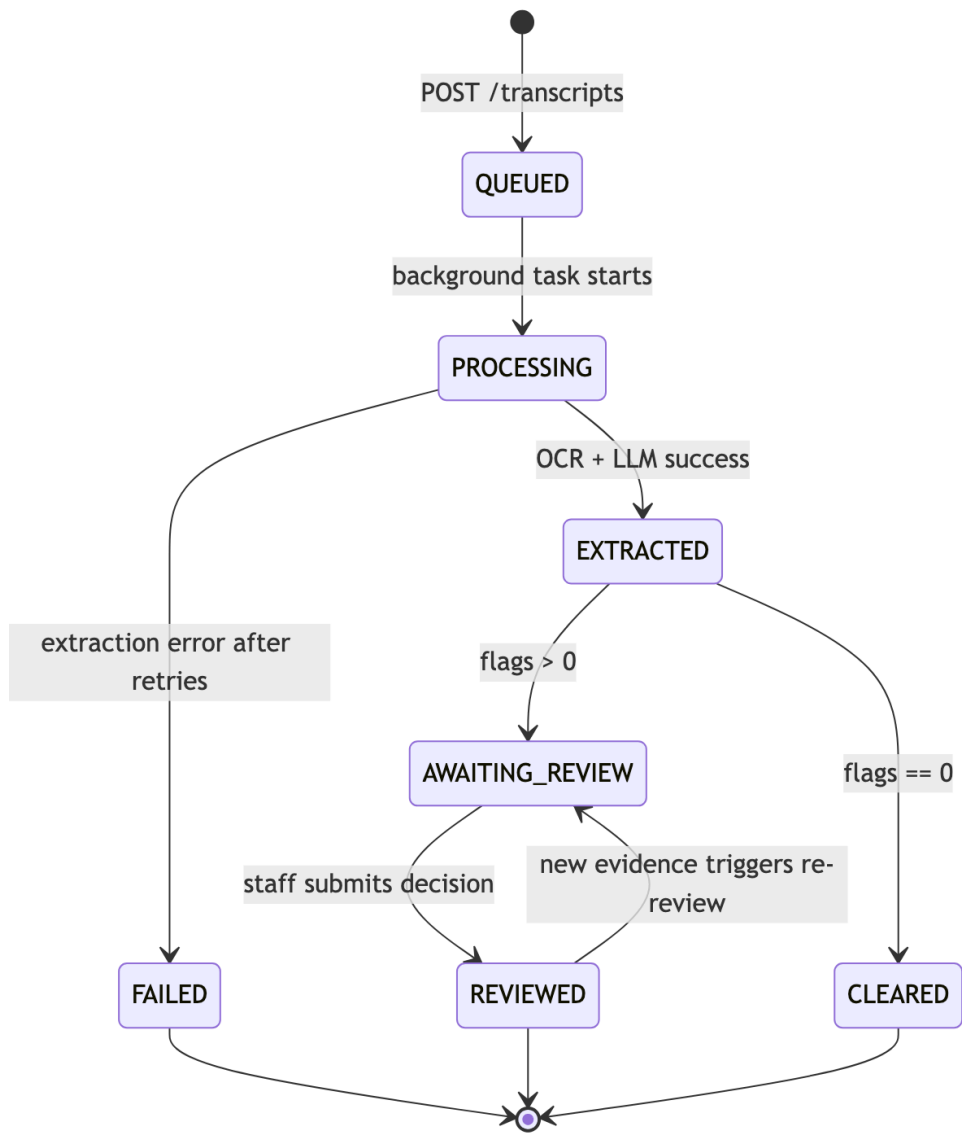


Srd Er Diagram

7.2 LLM-Returned JSON Schema (Extracted Data)

```
{
  "applicant_name": "string",
  "institution_name": "string",
  "program_name": "string",
  "graduation_date": "YYYY-MM-DD | null",
  "enrollment_date": "YYYY-MM-DD | null",
  "total_credits": "number | null",
  "courses": [
    {
      "code": "string",
      "title": "string",
      "credits": "number",
      "grade": "string"
    }
  ],
  "accreditation_body": "string | null",
  "raw_text_preview": "string (≤ 500 chars)"
}
```

7.3 Transcript State Machine



Srd State Machine

7.4 Data Retention

Entity	Retention	Notes
Original uploaded file	FILE_RETENTION_HOURS (default 24 h) after status reaches REVIEWED or CLEARED	Deleted by scheduled cleanup task
Transcript row + ExtractedData	Indefinite for the PoC	Phase B: configurable
VerificationFlag	Indefinite	Joined to transcript
StaffReview	Indefinite	Required for compliance
AuditLog	Indefinite, append-only	Required for audit per FR-082

8. System Constraints and Compliance

ID	Constraint / Compliance Item	Source
CMP-01	No automated licensure decisions	MSBON board policy
CMP-02	Synthetic / de-identified data only during development	FERPA
CMP-03	All data minimization principles followed (raw text not exported in audit)	NIST Privacy Framework
CMP-04	OWASP Top 10 (2021) addressed in design and implementation	NFR-SEC-05
CMP-05	WCAG 2.1 Level AA conformance	UI-003
CMP-06	LLM responses are advisory; staff retain decision authority	CMP-01
CMP-07	All deliverables under git version control with branch protection on main	CSC 424 tool requirement
CMP-08	Project teams shall record and submit meeting minutes	CSC 424 grading rubric
CMP-09	Final demonstration shall be a recorded video walkthrough	CSC 424 oral component

9. Acceptance Criteria and Traceability Matrix

9.1 Acceptance Criteria (Customer-Facing)

#	Acceptance Criterion	Target	Measurement Method
AC-01	Flag detection accuracy on the synthetic test set	≥ 90 %	Confusion matrix from tests/synthetic/
AC-02	Every flag includes rule ID + source excerpt + explanation	100 %	Schema-validated assertion
AC-03	Every state change produces an audit entry	100 %	pytest integration test
AC-04	Every OVERRIDDEN decision stores override_reason	100 %	DB constraint + service-layer check
AC-05	No endpoint approves or denies licensure	100 %	Code review + endpoint audit
AC-06	The end-to-end happy path is demonstrable in a recorded video	Pass / fail	Final presentation

9.2 Functional Requirement → Use Case → Test Traceability

Legend: ✓ = covered by automated test, △ = covered by manual demo step, — = pending Phase B/C.

Requirement	Use Case	Test Type	Status
FR-001..010 (Auth)	UC-01	Integration (tests/test_auth.py)	△
FR-020..029 (Upload)	UC-02	Integration + manual	△
FR-030..039 (Extraction)	UC-02	Integration + mocked LLM	—
FR-040..052 (Rules)	UC-02, UC-03	Unit (per rule)	△
FR-060..074 (Review)	UC-03	Integration	△
FR-080..089 (Audit)	UC-06	Integration	△
FR-090..099 (Programs)	UC-07	Integration	△
NFR-PERF-01..05	All	Manual stopwatch	△
NFR-SEC-01..10	All	Code review + pip-audit	△
NFR-USE-01..05	All	UAT	—

9.3 Demo Script (recorded presentation)

The capstone demonstration shall execute the following flow on screen (timing approximate, full script in §10.4):

1. **Login** as admin / demo (UC-01) → land on Transcript List.
2. Walk through **Programs** page; show seeded MS programs (UC-07).
3. **Upload** synthetic transcript A (clean BSN graduate) → watch status QUEUED → AWAITING_REVIEW; open it → 0–1 flags → CONFIRMED (UC-02, UC-03).
4. **Upload** synthetic transcript B (fraud indicators: 8-month BSN duration) → review flags → OVERRIDDEN with justification (UC-02, UC-03).
5. Open **Audit Log** → filter by transcript_id → demonstrate every action recorded (UC-06).
6. Click **Export CSV** → open the file in Numbers/Excel showing the audit trail (UC-05).

10. Appendix

10.1 Mapping to CSC 424 Capstone Deliverables

Capstone Deliverable	Where Addressed
Concept Document (10 %)	Original MSB0N Fraud-Sensitive Transcript

Capstone Deliverable	Where Addressed
Software Requirements Document (15 %)	Verification_Concept_CSC424_02.pdf This document
High Level Design Document (10 %)	Software_Design_Document.md (companion SDD)
High-Fidelity Mockup (10 %)	Live-rendered React UI on localhost:5173; screenshots in §10.3
Meeting Minutes (5 %)	docs/meeting-minutes/ directory
Documentation Quality (5 %)	This document + SDD + READMEs
Project Presentation (20 %)	Recorded walkthrough per §9.3

10.2 Flagging Rules Reference (canonical list — same as SDD §5.4)

Rule ID	Name	Category	Severity	Trigger
GRAD-001	Missing graduation confirmation	GRADUATION	HIGH	No explicit degree conferral text in raw transcript
GRAD-002	Graduation date absent	GRADUATION	MEDIUM	graduation_date field empty in extracted JSON
ACCR-001	Institution not in MS accredited list	ACCREDITATION	HIGH	institution_name not found in accredited_programs
ACCR-002	Accreditation body not recognized	ACCREDITATION	MEDIUM	accreditation_body $\notin$ {ACEN, CCNE}
COUR-001	Required nursing courses not present	COURSE	HIGH	One or more mandatory course codes missing
COUR-002	Insufficient credit hours	COURSE	MEDIUM	total_credits below minimum threshold (configurable)
FRAU-001	Program completion time below minimum	FRAUD	HIGH	(graduation_date - enrollment_date) < program minimum
FRAU-002	Suspicious grade distribution	FRAUD	MEDIUM	All-A pattern with no grade variance across $\geq$ 8 courses
FRAU-003	Institution on fraud watch list	FRAUD	HIGH	institution_name matches a known-fraud entry
FORM-001	Required transcript	FORMAT	LOW	One or more of {applicant, institution,

Rule ID	Name	Category	Severity	Trigger
	fields missing			program} could not be extracted
FORM-002	Inconsistent date formatting	FORMAT	LOW	Multiple date formats appear in raw text
FORM-003	Inconsistent grade formatting	FORMAT	LOW	Mixed letter / numeric / pass-fail grades

10.3 Glossary of Status Codes

Status	Meaning
QUEUED	Uploaded; awaiting background extraction
PROCESSING	Extraction in progress
EXTRACTED	LLM JSON persisted; rule engine pending
AWAITING_REVIEW	One or more flags raised; needs staff decision
CLEARED	Zero flags after verification
REVIEWED	Staff has submitted a decision
FAILED	Unrecoverable error during processing

10.4 Demo Script Detailed Steps

The full keystroke-level demo script is maintained in docs/DEMO_SCRIPT.md (created during Step 5 of the capstone implementation plan). It includes the synthetic transcript fixtures, expected flag set per fixture, and a 6-minute recording outline aligned with §9.3.

10.5 Open Issues / Future Work

#	Issue	Disposition
OI-1	Real user store (replace hardcoded demo users)	Phase B
OI-2	Field-level encryption for PII	Phase B (DB-010)
OI-3	Configurable rule authoring UI	Post-PoC
OI-4	PostgreSQL migration	Phase D (infrastructure decision required)
OI-5	Integration with MSBON production licensing DB	Out of capstone scope
OI-6	Mobile-responsive UI	Phase C
OI-7	E2E browser test suite (Playwright)	Phase C+

*End of Software Requirements Specification — MSBON Transcript Verification System
v1.0 PoC.*